

Transaction Fair Ordering for DAG-based BFT Consensus Protocols

Jérémie Decouchant¹

The order in which DeFi transactions run affects their outcome, which exposes users to costly reordering attacks. BFT consensus makes replicas agree on an order but places no constraint on how that order is chosen. Fair-ordering protocols close this gap, usually at a steep cost in performance. This talk will discuss the main ideas behind three recent results that reduce this cost using DAG-based consensus and/or trusted hardware:

- First, Tilikum [Se26] brings the two main fair-ordering properties, ordering linearizability and batch-order fairness, to DAG-based consensus without assuming weak edges.
- Next, Herring [PD26] speeds up batch-order fairness by building its ordering graph and processing committed sub-DAGs in parallel.
- Finally, Pearl asks whether trusted execution environments (TEEs) can reduce the number of replicas that fair ordering needs. Even with TEEs, more than $2f + 1$ replicas are required under partial synchrony. Under synchrony, however, $2f + 1$ replicas are enough, and Pearl shows how.

Along the way, the talk will also discuss the lessons we learned.

References

- [PD26] Putnik, M.; Decouchant, J.: Herring: Parallel Batch-Order-Fairness on DAG-based Blockchain Consensus, 2026, arXiv: 2605.23648 [cs.DC], URL: <https://arxiv.org/abs/2605.23648>.
- [Se26] Segalini, G.; Çolakoğlu, Y.; Putnik, M.; Decouchant, J.: Tilikum: Transaction Fair Ordering on a DAG without Weak Edges, 2026, arXiv: 2606.27250 [cs.CR], URL: <https://arxiv.org/abs/2606.27250>.

¹ Delft University of Technology, Department of Software Technology, Delft, The Netherlands, J.Decouchant@tudelft.nl